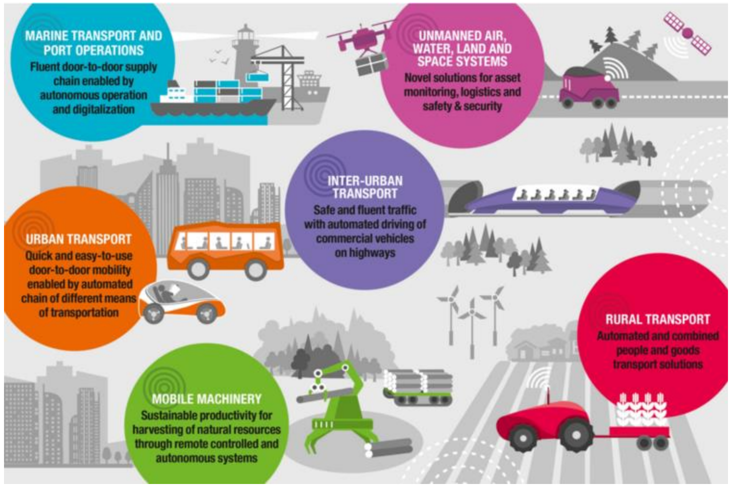
자율주행 시스템

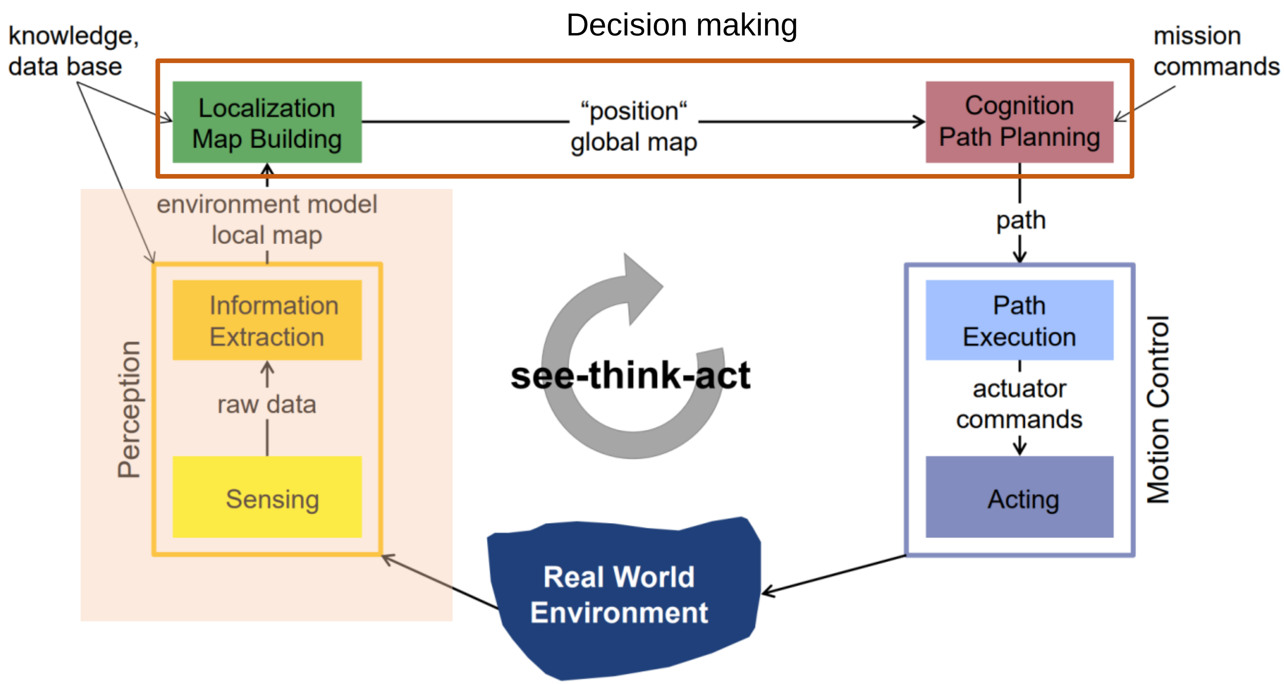
(Autonomous Driving System)

Lecture 10-1. Feature Extraction Using Range Data I



4. Perception

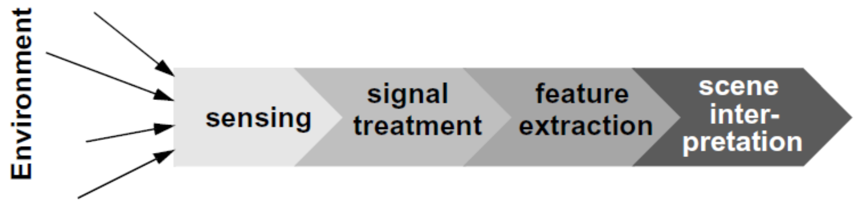
- **Autonomous Mobile Robot – the see-think-act cycle**



- ✓ One of the most important tasks of an autonomous system of any kind is to <u>acquire knowledge about its environment</u>.
- ✓ This is done by taking <u>measurement</u>s using various sensors and then <u>extracting meaningful information</u> from those measurements.

4.7 Feature Extraction Based on Range Data

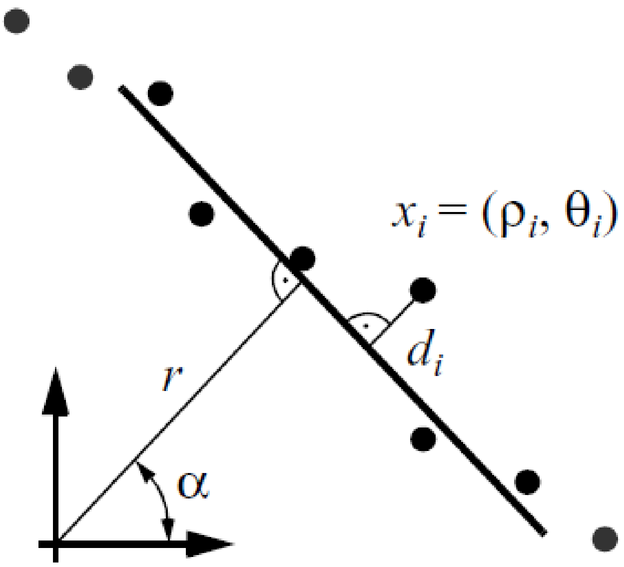
- **Feature extraction**
  - ✓ Reducing the number of resources required to describe a large set of data
  - ✓ A general term for methods of constructing combinations of the variables to get around these problems while still <u>describing the data with sufficient accuracy</u>.
  - → Extracting useful information from raw data



  - ✓ Line extraction: the simplest feature to extract
  - ✓ Three main problems in line extraction in unknown environments
    - How many lines are there?
    - Segmentation: Which points belong to which line?
    - Line fitting/Extraction: Given the points that belong to a line, how to estimate the line model parameters?

4.7.1 Line Fitting

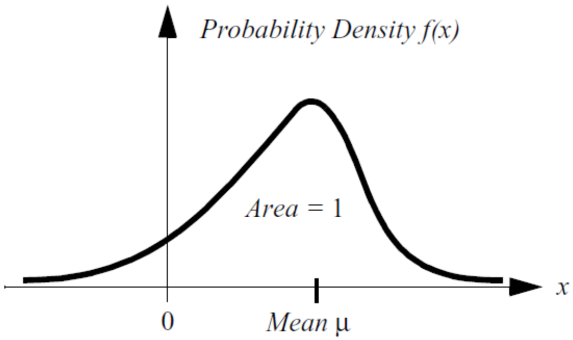
- **Probabilistic line fitting from uncertain range sensor data**
  - ✓ Goal of line fitting: To extract a line feature based on a set of sensor measurements



  - ✓ Uncertainty associated with each of the noisy range sensor measurements
    → no single line that passes through the set
  - ✓ Selection problem: to select the best possible match via optimization criterion.

4.7.1 Line Fitting

- **Probabilistic line fitting from uncertain range sensor data**
  - ✓ Probability Density Function (PDF) to characterize the statistical properties of a variable x



$$\int_{-\infty}^{\infty} f(x)dx = 1$$

  - ✓ Expected value of variable x:

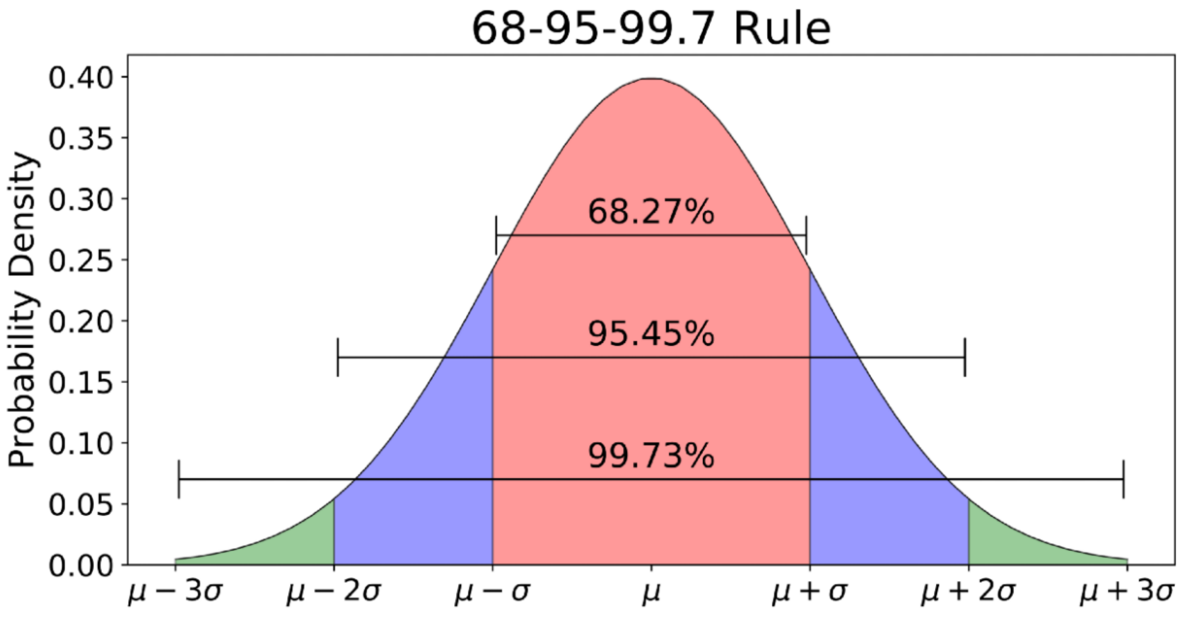
$$\mu = E[X] = \int_{-\infty}^{\infty} x f(x)dx$$

  - ✓ Variance of variable x:

$$\sigma^2 = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)dx$$

4.7.1 Line Fitting

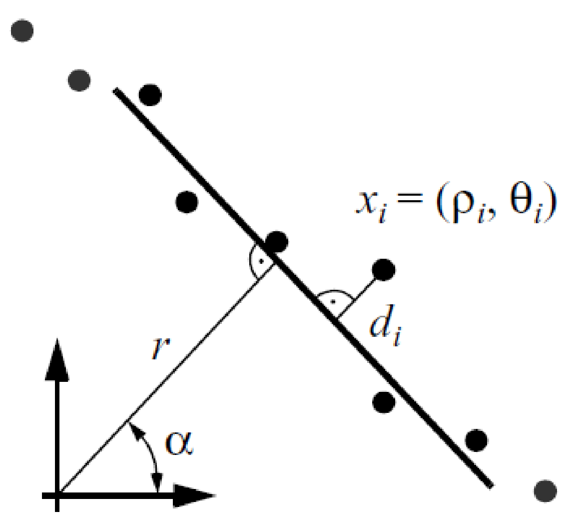
- **Probabilistic line fitting from uncertain range sensor data**
  - ✓ Most common PDF for characterizing uncertainties: Gaussian (normal) distribution

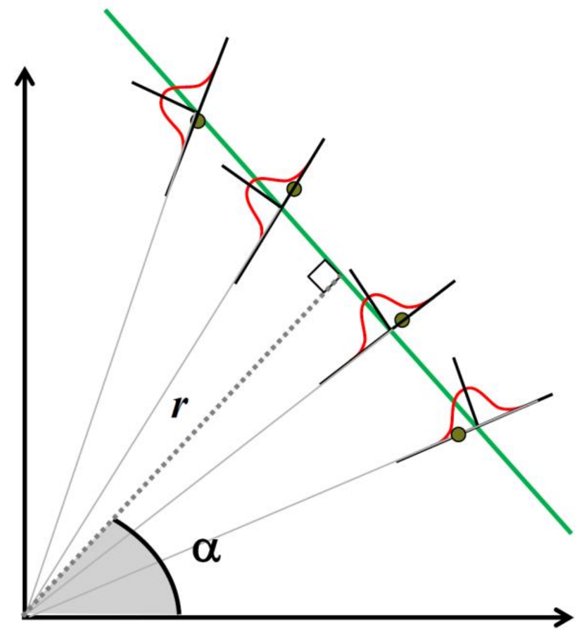


$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

4.7.1 Line Fitting

- **Probabilistic line fitting from uncertain range sensor data**
  - ✓ Imagine extracting a line based on point measurements with uncertainties.
  - ✓ Model parameters in polar coordinates [(r, α) uniquely identifies a line]
  - ✓ What is the uncertainty of the extracted line knowing the uncertainties of the measurement points that contribute to it ?

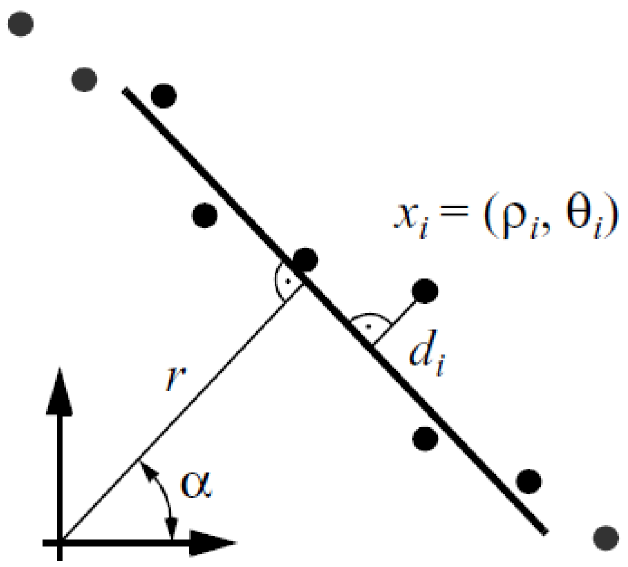




4.7.1 Line Fitting

- **Probabilistic line fitting from uncertain range sensor data**
  - ✓ Given some measurement point (ρ, θ),

$$x = \rho \cos \theta \, , y = \rho \sin \theta$$



4.7.1 Line Fitting

- **Probabilistic line fitting from uncertain range sensor data**
  - ✓ Point-Line distance

$$\rho_i \cos(\theta_i - \alpha) - r = d_i$$

  - ✓ If each measurement is equally uncertain then sum of sq. errors:

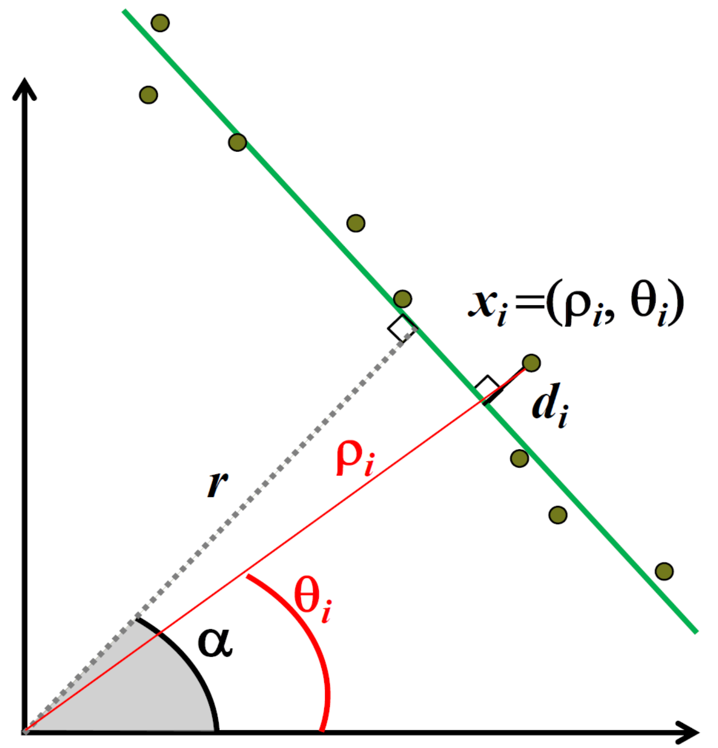
$$S = \sum_i d_i^2 = \sum_i (\rho_i \cos(\theta_i - \alpha) - r)^2$$

  - ✓ Goal: minimize S when selecting (r, α)

    → a solve the system

$$\frac{\partial S}{\partial \alpha} = 0, \qquad \frac{\partial S}{\partial r} = 0$$

  - ✓ "Unweighted Least Squares"



4.7.1 Line Fitting

- **Probabilistic line fitting from uncertain range sensor data**
  - ✓ Point-Line distance

$$\rho_i \cos(\theta_i - \alpha) - r = d_i$$

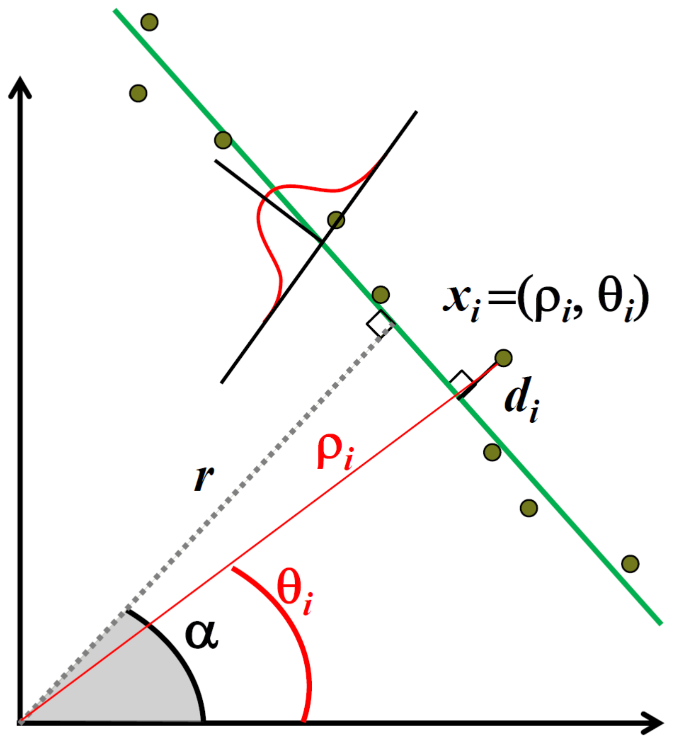
  - ✓ Each sensor measurement, may have its own, unique uncertainty, σ_i^2

$$S = \sum_i w_i d_i^2 = \sum_i w_i (\rho_i \cos(\theta_i - \alpha) - r)^2$$

$$w_i = 1/\sigma_i^2$$

  - ✓ “Weighted Least Squares”

$$\frac{\partial S}{\partial \alpha} = 0, \qquad \frac{\partial S}{\partial r} = 0$$



4.7.1 Line Fitting

- **Probabilistic line fitting from uncertain range sensor data**
  - ✓ Weighted least squares and solving the system:

$$\frac{\partial S}{\partial \alpha} = 0, \qquad \frac{\partial S}{\partial r} = 0$$

  - ✓ Gives the line parameters

$$\alpha = \frac{1}{2} atan\left(\frac{\sum w_i \rho_i^2 \cos 2\theta_i - \frac{2}{\sum w_i}\sum\sum w_i w_j\, \rho_i \rho_j cos\theta_i sin\theta_j}{\sum w_i \rho_i^2 \cos 2\theta_i - \frac{1}{\sum w_i}\sum\sum w_i w_j\, \rho_i \rho_j cos\left(\theta_i + \theta_j\right)}\right)$$

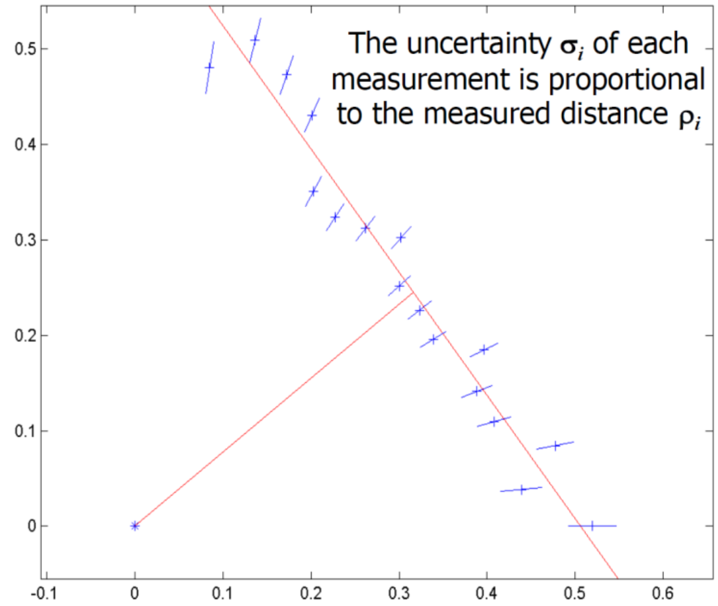
$$r = \frac{\sum w_i \rho_i \cos(\theta_i - \alpha)}{\sum w_i}$$

If $\begin{cases} \rho_i \sim N\left(\widehat{\rho}_i, \sigma_{\rho_i}^2\right) \\ \theta_i \sim N\left(\widehat{\theta}_i, \sigma_{\theta_i}^2\right) \end{cases}$, what is the uncertainty in the line (r, α) ?

4.7.1 Line Fitting

- **Probabilistic line fitting from uncertain range sensor data**

Pointing angle of sensor θ_i [deg]	Range θ_i [m]	Pointing angle of sensor θ_i [deg]	Range θ_i [m]
0	0.5197	45	0.4276
5	0.4404	50	0.4075
10	0.485	55	0.3956
15	0.4222	60	0.4053
20	0.4132	65	0.4752
25	0.4371	70	0.5032
30	0.3912	75	0.5273
35	0.3949	80	0.4879
40	0.3919		



$$S = \sum_i d_i^2 = \sum_i (\rho_i \cos(\theta_i - \alpha) - r)^2$$

✓ Goal: minimize S when selecting (r, α)

$$\frac{\partial S}{\partial \alpha} = 0, \qquad \frac{\partial S}{\partial r} = 0$$

Answer:

$\alpha = 37.36\ °$

$R\ =\ 0.4$